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A

## ***B.Tech. Degree VI Semester Regular Examination April 2022***

### **ME 19-205-0601 DYNAMICS OF MACHINERY (2019 Scheme)**

Time: 3 Hours

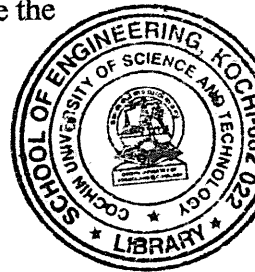
Maximum Marks: 60

**Course Outcome**

On successful completion of the course, the students will be able to:

- CO1: Recall the concepts of free body diagrams, principles of statics and dynamics  
 CO2: Use graphical and analytic methods to do force analysis of planar mechanisms  
 CO3: Apply these concepts in different machine elements for the evaluation of forces and moments  
 CO4: Analyze the dynamics of different mechanisms and machine elements and determine the various forces and torques.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create  
 PO – Programme Outcome



### **PART A (Answer ALL questions)**

|    |     | (8 × 3 = 24)                                                                                                                                                                                                                    | Marks | BL | CO | PO |
|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| I. | (a) | Explain the conditions for a body to be in equilibrium under the action of two forces, three forces and two force and torque.                                                                                                   | 3     | L2 | 1  | 1  |
|    | (b) | The inertia of the connecting rod can be replaced by two masses concentrated at two points and connected rigidly together. How to determine the two masses so that it is dynamically equivalent to the connecting rod? Justify: | 3     | L1 | 1  | 2  |
|    | (c) | Explain the turning moment diagram of a multicylinder engine.                                                                                                                                                                   | 3     | L2 | 4  | 1  |
|    | (d) | Explain the concept of gyroscopic couple? Derive the formula for its magnitude.                                                                                                                                                 | 3     | L2 | 4  | 2  |
|    | (e) | Define the terms 'static balancing' and 'dynamic balancing'. State the necessary conditions to achieve them.                                                                                                                    | 3     | L1 | 3  | 1  |
|    | (f) | Derive the following expressions, for an uncoupled two cylinder locomotive engine:                                                                                                                                              | 3     | L1 | 4  | 2  |
|    |     | (i) Variation is tractive force                                                                                                                                                                                                 |       |    |    |    |
|    |     | (ii) Swaying couple                                                                                                                                                                                                             |       |    |    |    |
|    |     | (iii) Hammer blow                                                                                                                                                                                                               |       |    |    |    |
|    | (g) | Show that the speed at which a belt or rope should be run to transmit maximum power is that at which the maximum allowable tension is three times the centrifugal tension in the belt or rope at that speed.                    | 3     | L1 | 4  | 3  |
|    | (h) | Explain stability of a governor? Sketch the controlling force versus radius diagrams for a stable, unstable and isochronous governor.                                                                                           | 3     | L2 | 4  | 2  |

### **PART B**

(4 × 12 = 48)

|     |                                                                                                                                                                                                                                                                          |    |    |   |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| II. | Determine the required input torque on the crank of a Slider-crank mechanism for static equilibrium when the applied piston load is 1500 N. The length of crank and connecting rod are 40 mm and 100 mm respectively and the crank is turned 45° from inner dead centre. | 12 | L3 | 2 | 2 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|

OR

(P.T.O.)

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- III. A horizontal steam engine running at 120 r.p.m. has a bore of 250 mm and a stroke of 400 mm. The connecting rod is 0.6 m and mass of the reciprocating parts is 60 kg. When the crank has turned through an angle of  $45^\circ$  from the inner dead Centre, the steam pressure on the cover end side is  $550 \text{ kN/m}^2$  and that on the crank end side is  $70 \text{ kN/m}^2$ . Considering the diameter of the piston rod equal to 50 mm, determine:
- turning moment on the crank shaft
  - thrust on the bearings
  - acceleration of the flywheel, if the power of the engine is 20 kW, mass of the flywheel 60 kg and radius of gyration 0.6 m.
- 12 L3 1 2
- IV. The turning moment diagram of a four stroke engine may be assumed for the sake of simplicity to be represented by four triangles in each stroke. The areas of these triangles are as follows:
- Suction stroke =  $5 \times 10^{-5} \text{ m}^2$ ; Compression stroke =  $21 \times 10^{-5} \text{ m}^2$ ;  
Expansion stroke =  $85 \times 10^{-5} \text{ m}^2$ ; Exhaust stroke =  $8 \times 10^{-5} \text{ m}^2$
- All the areas excepting expansion stroke are negative. Each  $\text{m}^2$  of area represents 14 MN-m of work. Assuming the resisting torque to be constant, determine the moment of inertia of the flywheel to keep the speed between 98 r.p.m. and 102 r.p.m. Also find the mass of a rim-type flywheel, given that density of flywheel material is  $8150 \text{ kg/m}^3$ ; the allowable tensile stress of the flywheel material is 7.5 MPa.
- 12 L3 4 2
- OR
- V. A ship propelled by a turbine rotor which has a mass of 5 tonnes and a speed of 2100 r.p.m. The rotor has a radius of gyration of 0.5 m and rotates in a clockwise direction when viewed from the stern. Find the gyroscopic effects in the following conditions:
- The ship sails at a speed of 30 km/h and steers to the left in a curve having 60 m radius.
  - The ship pitches 6 degree above and 6 degree below the horizontal position. The bow is descending with its maximum velocity. The motion due to pitching is simple harmonic and the periodic time is 20 seconds.
  - The ship rolls and at a certain instant it has an angular velocity of 0.03 rad/s clockwise when viewed from stern.
- Determine the maximum angular acceleration during pitching. Explain how the direction of motion due to gyroscopic effect is determined in each case.
- 12 L3 4 2

(Continued)

- VI. A shaft carries four masses in parallel planes A, B, C and D in this order along its length. The masses at B and C are 18 kg and 12.5 kg respectively, and each has an eccentricity of 60 mm. The masses at A and D have an eccentricity of 80 mm. The angle between the masses at B and C is  $100^\circ$  and that between the masses at B and A is  $190^\circ$ , both being measured in the same direction. The axial distance between the planes A and B is 100 mm and that between B and C is 200 mm. If the shaft is in complete dynamic balance, determine:
- the magnitude of the masses at A and D
  - the distance between planes A and D
  - the angular position of the mass at D.
- OR**
- VII. The following data apply to an outside cylinder uncoupled locomotive:
- Mass of rotating parts per cylinder = 360 kg;  
 Mass of reciprocating parts per cylinder = 300 kg ;  
 Angle between cranks =  $90^\circ$  ;  
 Crank radius = 0.3 m ;  
 Cylinder centres = 1.75 m ;  
 Radius of balance masses = 0.75 m ;  
 Wheel centres = 1.45 m.
- If whole of the rotating and two-thirds of reciprocating parts are to be balanced in planes of the driving wheels, find:
- magnitude and angular positions of balance masses
  - speed in kilometres per hour at which the wheel will lift off the rails when the load on each driving wheel is 30 kN and the diameter of tread of driving wheels is 1.8 m
  - swaying couple at speed arrived at in above case.
- VIII. The following data refer to an open belt drive:
- Diameter of larger pulley = 400 mm ;  
 Diameter of smaller pulley = 250 mm ;  
 Distance between two pulleys = 2 m ;  
 Coefficient of friction between smaller pulley surface and belt = 0.4 ;  
 Maximum tension when the belt is on the point of slipping = 1200 N.  
 Find the power transmitted at speed of 10 m/s. It is desired to increase the power. Which of the following two methods you will choose?
- Increasing the initial tension in the belt by 10 per cent
  - Increasing the coefficient of friction between the smaller pulley surface and belt by 10 per cent by the application of suitable dressing on the belt.
- Also find the percentage increase in power possible in each case.
- OR**
- IX. A Porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 25 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Find the range of speed, sleeve lift, governor effort and power of the governor in the following cases:
- when the friction at the sleeve is neglected
  - when the friction at the sleeve is equivalent to 10 N.

Bloom's Taxonomy Levels

L1-25 , L2-25%, L3-43.75%, L4-6.25%, L5-0%